

DOULAN, China

WMO: 528360

Lat: 36.2958N

Lon: 98.0964E

Elev: 3190

StdP: 68.43

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-16.0	-31.1	0.3	-8.6	-29.4	0.4	-7.8	7.6	-5.4	7.1	-4.6	1.9	140	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	25.8	11.3	23.9	10.3	22.3	9.5	13.2	21.8	12.2	20.2	11.4	18.8	2.4	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
10.5	11.7	15.6	9.5	11.0	15.0	8.5	10.3	14.3	48.8	21.8	45.6	20.3	42.8	18.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.3	6.3	5.5	-21.0	29.0	2.4	1.5	-22.7	30.0	-24.1	30.9	-25.4	31.8	-27.2	32.9
			-21.8	14.6	2.3	1.3	-23.5	15.5	-24.8	16.3	-26.0	17.0	-27.7	17.9
				</										

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-8.3	-4.7	-0.1	5.4	9.6	13.0	15.5	15.3	10.9	4.4	-1.9	-7.1	(6)
(7)		DBStd	8.98	3.73	4.09	3.81	3.73	3.18	2.91	3.18	2.93	3.08	3.33	3.71	3.32	(7)
(8)		HDD10.0	2583	568	413	315	145	47	8	1	1	23	176	357	531	(8)
(9)		HDD18.3	5119	827	646	573	388	272	162	98	104	222	432	607	789	(9)
(10)		CDD10.0	527	0	0	1	7	33	97	173	163	51	3	0	0	(10)
(11)		CDD18.3	22	0	0	0	0	0	1	12	8	1	0	0	0	(11)
(12)		CDH23.3	218	0	0	0	1	1	14	112	81	10	0	0	0	(12)
(13)		CDH26.7	27	0	0	0	0	0	1	16	9	1	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.4	2.6	2.9	2.9	2.7	2.4	2.2	2.1	2.3	2.4	2.4	(14)	
(15)	Precipitation	PrecAvg	206	4	6	7	9	28	47	47	29	18	7	4	3	(15)
(16)		PrecMax	350	11	21	27	39	79	142	109	93	48	28	17	13	(16)
(17)		PrecMin	107	0	0	0	0	0	4	11	4	0	0	0	0	(17)
(18)		PrecStd	60	3	5	6	10	18	26	26	18	12	6	4	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	10.0	15.4	19.9	22.5	25.1	28.6	27.9	24.7	18.5	11.1	6.3	(19)
(20)			MCWB	-2.9	-0.6	2.2	5.0	6.7	10.6	12.6	12.6	11.1	5.0	0.2	-2.6	(20)
(21)		2%	DB	2.3	7.0	11.2	17.1	20.2	23.0	26.2	25.6	21.4	15.1	8.5	3.0	(21)
(22)			MCWB	-4.8	-2.1	0.1	3.5	5.7	9.1	11.8	11.1	8.9	3.2	-1.0	-4.3	(22)
(23)		5%	DB	0.0	4.4	8.9	15.0	18.4	21.3	24.2	23.6	19.2	12.8	6.4	0.8	(23)
(24)			MCWB	-6.1	-3.4	-1.0	2.6	5.0	8.1	10.7	10.3	7.6	2.1	-2.1	-5.6	(24)
(25)		10%	DB	-2.1	2.1	6.7	12.8	16.5	19.8	22.5	21.9	17.3	10.9	4.5	-1.1	(25)
(26)			MCWB	-7.4	-4.6	-2.0	1.6	4.3	7.4	10.2	9.5	6.4	1.3	-3.1	-6.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	-0.5	2.4	5.6	8.2	13.1	14.7	14.5	12.6	6.6	0.6	-2.4	(27)
(28)			MCDWB	5.1	9.1	14.4	17.3	16.9	21.7	24.4	23.2	20.9	15.5	10.3	5.8	(28)
(29)		2%	WB	-4.6	-2.0	0.5	4.2	7.1	11.2	13.5	13.0	10.8	4.4	-0.8	-4.3	(29)
(30)			MCDWB	1.9	6.3	10.2	15.4	15.7	18.6	22.3	21.7	17.7	12.3	7.7	2.6	(30)
(31)		5%	WB	-6.0	-3.2	-0.7	3.1	6.2	10.1	12.4	12.1	9.4	3.1	-1.9	-5.5	(31)
(32)			MCDWB	-0.3	4.1	7.9	13.7	14.9	16.9	20.5	20.2	16.4	11.1	5.8	0.5	(32)
(33)		10%	WB	-7.2	-4.5	-1.8	2.0	5.4	9.1	11.5	11.2	8.1	1.8	-2.9	-6.6	(33)
(34)			MCDWB	-2.3	1.7	6.1	11.8	13.9	16.1	19.0	18.6	15.0	9.8	3.9	-1.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.8	10.8	10.7	11.2	10.9	10.4	10.4	11.1	11.2	11.1	10.8	10.8	(35)
(36)			MCDBR	12.9	13.5	13.5	14.0	14.1	13.3	13.5	13.7	13.7	13.3	12.6	12.7	(36)
(37)			MCWBR	8.3	7.9	6.9	6.2	5.7	4.8	4.8	4.8	5.4	6.0	6.8	8.0	(37)
(38)		5% WB	MCDBR	12.6	12.8	12.5	12.9	11.1	10.3	10.7	11.0	11.2	11.7	12.1	12.4	(38)
(39)			MCWBR	8.2	7.5	6.5	5.8	4.6	4.2	4.3	4.1	4.6	5.5	6.6	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.219	0.257	0.311	0.360	0.371	0.362	0.350	0.340	0.315	0.274	0.233	0.214	(40)	
(41)		taud	2.477	2.367	2.173	2.042	2.099	2.229	2.346	2.360	2.399	2.432	2.470	2.499	(41)	
(42)		Ebn at Noon	1010	993	954	919	911	915	924	930	941	960	980	998	(42)	
(43)		Edh at Noon	80	102	137	166	160	140	124	120	109	95	80	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.12	3.93	5.25	6.42	6.77	6.56	6.59	6.24	5.33	4.60	3.53	2.94	(44)	
(45)		RadStd	0.23	0.23	0.31	0.30	0.39	0.49	0.46	0.44	0.37	0.20	0.19	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page